MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MIL-STD-188C/RS232C ASYNCHRONOUS SERIAL CONTROL MODULE

KEY CODE AEK

This amendment forms a part of Military Specification MIL-M-28787/521, dated 20 September 1989, and is approved for use by the Crane Division, Naval Surface Warfare Center, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGES 4 THROUGH 6

TABLE I: Delete section of table I and substitute new section of table I as printed on pages 3 through 5 of this amendment.

PAGE 24

6.3.2, Output parameters, Signal IOCA REQ, Ios max (mA) column: Delete "-15.0" and substitute "-30.0".

PAGE 25

6.3.2, Output parameters, Signal CHAN PRES, Iozh max (mA) column: Delete "1.0" and substitute "0.25".

PAGES 52 AND 53

TABLE VI, Test conditions E4, E8-E10, E14, E15, E17-E21, E23, E28, E31 and E32, Min limit: Delete "-260.0" (fifteen places) and substitute "-220.0" (fifteen places).

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TABLE VI, Test condition E14, Max limit: Delete "-15.0" and substitute "-30.0".

AMSC N/A FSC 5963
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TABLE VI, Test tolerance table, Output current measurement, Accuracy column, Test conditions E4, E8-E10, E15, E17-E21, E23, E28, E31 and E32: Delete "11.5" (seven places) and substitute "9.5" (seven places).

TABLE VI, Test tolerance table, Output current measurement, Accuracy column, Test condition E14: Delete "12.2" and substitute "9.5".

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TABLE XI, Test condition K5, Min limit: Delete "-1.0" and substitute "-0.02".

TABLE XI, Test condition K5, Max limit: Delete "1.0" and substitute "0.25".

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TABLE XI, Test tolerance table, Output current measurement, Accuracy column, Test condition K5: Delete "100" and substitute "13.5".

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FIGURE 11: Delete and substitute new figure 11 as printed on page 6 of this amendment.

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FIGURE 12a: Delete and substitute new figure 12a as printed on page 7 of this amendment.

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FIGURE 12c: Delete and substitute new figure 12c as printed on page 8 of this amendment.

Preparing activity:
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TABLE I. Module electrical requirements.

Test cond.	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
	End-of-life	II	S	1	INF	1	INF	1	INF	MOhm
A2-A6	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
	End-of-life	II	X	1	INF	1	INF	1	INF	MOhm
B1-B8	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	---------Module shall pass all remaining tests---------						
D1	Ibd	V	S	-0.12	1.21	-0.12	1.21	-0.12	1.21	mA
D2-D4	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D5	Ibd	V	S	-0.17	1.27	-0.17	1.27	-0.17	1.27	mA
D6	Ibd	V	S	-0.12	0.22	-0.12	0.22	-0.12	0.22	mA
D7	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D8	Ibd	V	S	-0.12	1.11	-0.12	1.11	-0.12	1.11	mA
D9-D15	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D16,D17	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D18	Ibd	V	S	0.0	4.0	0.0	4.0	0.0	4.0	mA
D19-D23	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D24	Ibd	V	S	-0.12	0.72	-0.12	0.72	-0.12	0.72	mA
D25	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D26	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D27	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
E1-E3	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
E4	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E5-E7	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
E8-E10	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E11-E13	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
E14,E15	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E16	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
E17-E21	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E22	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
E23	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E24-E27	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond.	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
E28	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E29	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
E30	Ios	VI	S	-7.84	-5.18	-7.84	-5.18	-7.84	-5.18	mA
E31,E32	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E33	Ios	VI	S	-85.0	-15.0	-85.0	-15.0	-85.0	-15.0	mA
E34-E40	Ios	VI	S	-225.0	-40.0	-225.0	-40.0	-225.0	-40.0	mA
E41	Ios	VI	S	-85.0	-15.0	-85.0	-15.0	-85.0	-15.0	mA
E42-E57	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
F1	Iih	VII	X	-5.88	-2.84	-5.88	-2.84	-5.88	-2.84	mA
F2-F4	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F5	Iih	VII	X	-15.2	-4.11	-15.2	-4.11	-15.2	-4.11	mA
F6	Iih	VII	X	-5.63	-2.85	-5.63	-2.85	-5.63	-2.85	mA
F7	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F8	Iih	VII	X	-7.43	-2.82	-7.43	-2.82	-7.43	-2.82	mA
F9-F13	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F14	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F15	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F16,F17	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F18	Iih	VII	X	-8.0	0.2	-8.0	0.2	-8.0	0.2	mA
F19-F23	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F24	Iih	VII	X	-6.83	-2.75	-6.83	-2.75	-6.83	-2.75	mA
F25	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F26	Iih	VII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
F27-F42	Iih	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
F43	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G1	Iil	VIII	X	-5.88	-2.84	-5.88	-2.84	-5.88	-2.84	mA
G2-G4	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G5	Iil	VIII	X	-15.2	-4.11	-15.2	-4.11	-15.2	-4.11	mA
G6	Iil	VIII	X	-5.63	-2.85	-5.63	-2.85	-5.63	-2.85	mA
G7	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G8	Iil	VIII	X	-7.43	-2.82	-7.43	-2.82	-7.43	-2.82	mA
G9-G13	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond.	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G14	IIi	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G15	IIi	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G16,G17	IIi	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G18	IIi	VIII	X	-8.0	0.2	-8.0	0.2	-8.0	0.2	mA
G19-G23	IIi	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G24	IIi	VIII	X	-6.83	-2.75	-6.83	-2.75	-6.83	-2.75	mA
G25	IIi	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G26	IIi	VIII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
G27-G42	IIi	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G43	IIi	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
H1-H56	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J58	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K4	Iozh	XI	X	-0.02	0.02	-0.02	0.02	-0.02	0.02	mA
K5	Iozh	XI	X	-0.02	0.25	-0.02	0.25	-0.02	0.25	mA
K6-K9	Iozh	XI	X	-0.02	0.02	-0.02	0.02	-0.02	0.02	mA
K10-K17	Iozl	XI	X	-0.02	0.02	-0.02	0.02	-0.02	0.02	mA
L1-L4	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L5-L9	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L10	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L11-L14	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L15-L22	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
M	PSFC	XIII	S	5.4	10.0	5.4	10.0	5.4	10.0	µF
N	Icc	XIV	X	500	1500	500	1500	500	1500	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						

- See footnotes at end of table.

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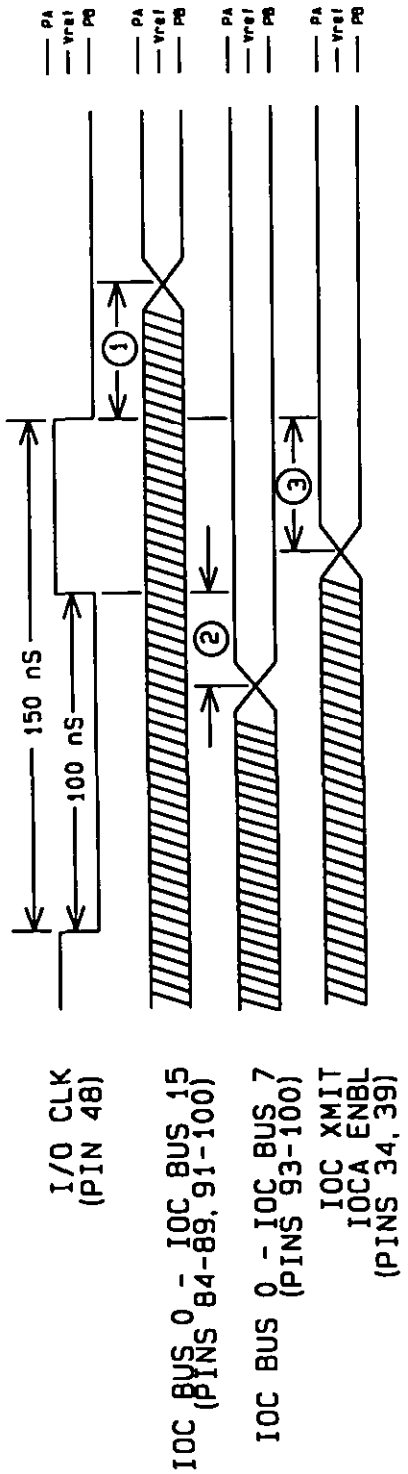


FIGURE 11. Input timing parameters for table XV.

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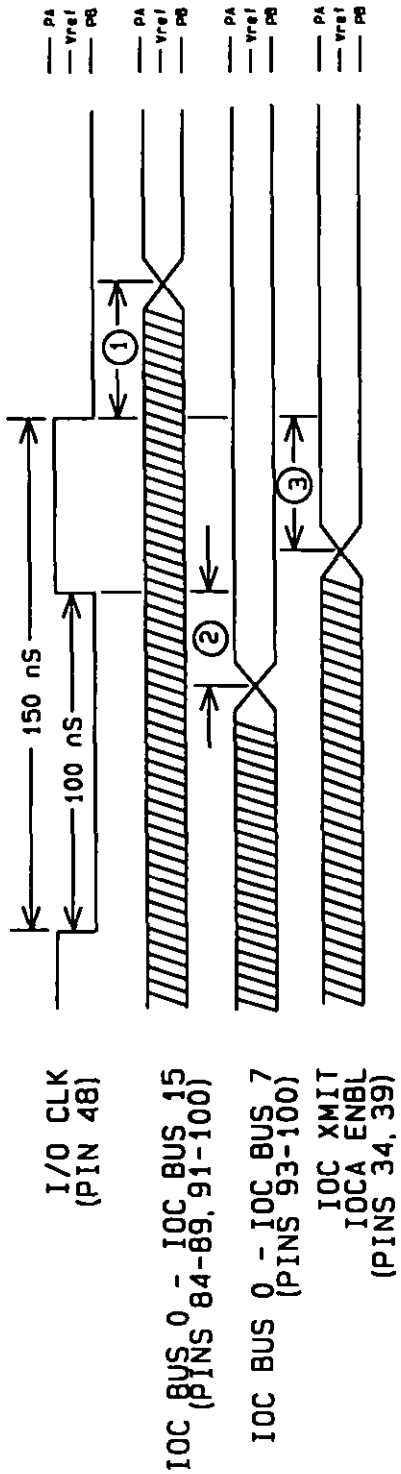


FIGURE 12a. Dynamic waveforms for table XVII.

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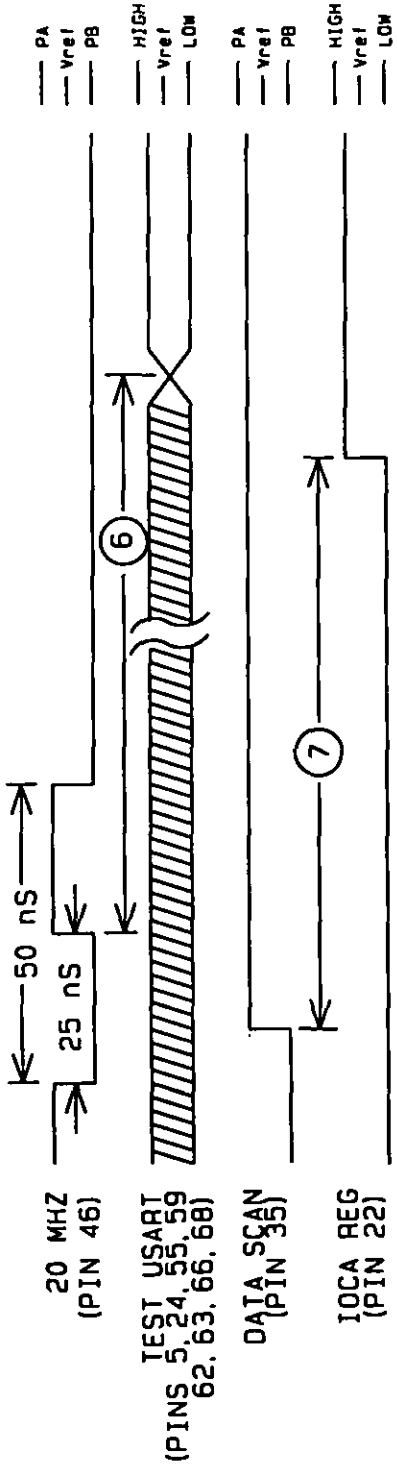


FIGURE 12c. Dynamic waveforms for table XVII.

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